

# Burmese Grapheme–Phoneme SMT with Moses & GIZA

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# Overview

# 2 preprocessing – 2 data – 2 sgms – 2 configs

- Two approaches of preprocessing (.my) data
- 1 = **data/cleaned/ .. & test-sgm = config.baseline**
- 2 = **data/cleaned\_2/ .. & test-sgm = config.baseline2**

# 8 runs – 4 notebooks – 2 baseline folders

- **pbsmt\_v1.ipynb** = run1, run2 = **baseline/../../steps/ 1 & 2**
- **pbsmt\_v2.ipynb** = run3 = **baseline/../../steps/3**
- **pbsmt\_v3.ipynb** = run4, run5 = **baseline2/../../steps/ 1 & 2**
- **pbsmt\_v4.ipynb** = run6, run7, run8 = **baseline2/../../steps/3**

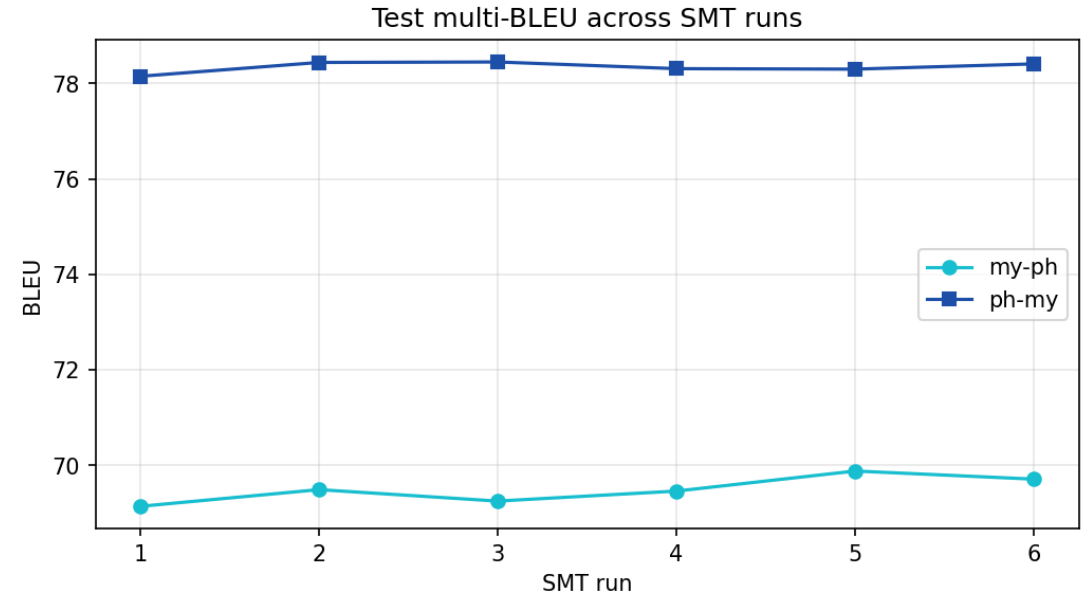
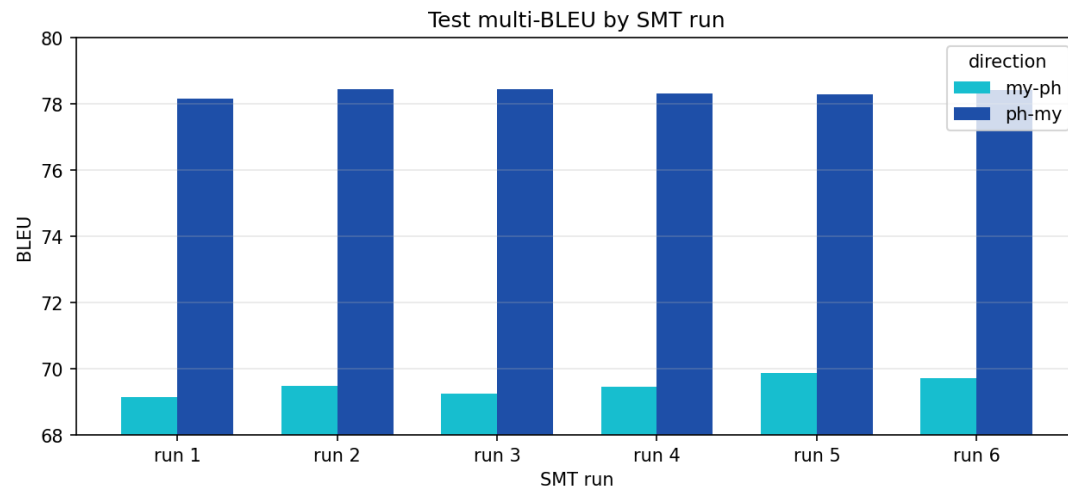
# 6 end-to-end SMTs – 8 runs

- A total of 6 successful/complete end-to-end SMT runs:
  1. Run1 + run2 + run3 (5-gram, **data/cleaned**, --prune '0 0 1', GIZA)
  2. Run4 (5-gram, **data/cleaned\_2**, --prune '0 0 1', GIZA)
  3. Run5 (3-gram, **data/cleaned\_2**, --prune '0 0 1', GIZA)
  4. Run6 (3-gram, **data/cleaned\_2**, --prune '0 0 1', GIZA)
  5. Run7 (3-gram, **data/cleaned\_2**, --prune '0 0 0', GIZA)
  6. Run8 (3-gram, **data/cleaned\_2**, --prune '0 0 0', MGIZA)

# Summary of Multi-BLEU scores

- 1st successful/complete end-to-end SMT (run1 + run2 + run3):
  - (my-ph): BLEU = 69.14, 85.2/72.4/64.3/57.6 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.15, 87.8/79.4/75.0/71.3 (BP=1.000, ratio=1.000, hyp\_len=8047, ref\_len=8048)
- 2nd end-to-end SMT (run4), create new data with updated syllable normalization:
  - (my-ph): BLEU = 69.49, 85.2/72.6/64.8/58.2 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.44, 87.8/79.6/75.6/71.7 (BP=1.000, ratio=1.000, hyp\_len=8046, ref\_len=8048)
- 3rd end-to-end SMT (run5), reduce order from 5-gram to 3-gram:
  - (my-ph): BLEU = 69.25, 85.3/72.7/64.7/57.4 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.45, 87.9/79.7/75.6/71.7 (BP=1.000, ratio=1.000, hyp\_len=8046, ref\_len=8048)
- 4th end-to-end SMT (run6), change max-phrase-length from 5 to 3:
  - (my-ph): BLEU = 69.46, 85.1/72.5/64.2/58.8 (BP=1.000, ratio=1.000, hyp\_len=8051, ref\_len=8048)
  - (ph-my): BLEU = 78.31, 87.9/79.6/75.4/71.4 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)
- 5th end-to-end SMT (run7), less prune:
  - (my-ph): BLEU = 69.88, 85.3/72.8/65.0/59.1 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.30, 87.7/79.3/75.4/71.8 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)
- 6th end-to-end SMT (run8), use MGIZA instead of GIZA:
  - (my-ph): BLEU = 69.71, 85.1/72.7/65.0/58.7 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.41, 87.8/79.5/75.5/71.8 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)

# Summary of Multi-BLEU scores



# Details

# Data Preprocessing

- Move assignment data files under **data/g2p-par/**
- Syllable-normalization with **syl\_normalizer.py** (Version 0.6) on **data/g2p-par/train.my**, **../dev.my**, and **../test.my**
- Create **data/cleaned** for normalized .my files

# Run 1 (SMT run 1)

- Modify paths in **config.baseline** & Perl scripts
- Generate configs & SGMs
- Run
- Error
- Add missing "**mkcls**" to **giza-pp/GIZA++-v2/**

# Run 2 (SMT run 1)

- Run again
- Error: .ps & .png files are corrupted
- Install ImageMagick & Graphviz to Ubuntu
- Manually convert .dot files to .ps & .png formats
- Multi-BLEU score:
  - (my-ph): BLEU = 69.14, 85.2/72.4/64.3/57.6 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.15, 87.8/79.4/75.0/71.3 (BP=1.000, ratio=1.000, hyp\_len=8047, ref\_len=8048)

# Run 2 (SMT run 1)

- Error: NIST-BLEU & NIST-BLEU C are not computed
  - Can't find Unicode property definition "Line\_Break" in regex; marked by <-  
- HERE in m/\p{Line\_Break} <-- HERE \p{Zl}/ at  
/home/lawun330/NLP/ubuntu-17.04/moses/scripts/generic/mteval-  
v13a.pl line 950.
- Edit **ubuntu-17.04/moses/scripts/generic/mteval-v13a.pl** to  
use \p{Line\_Break=Hyphen} instead of \p{Line\_Break}

# Run 3 (SMT run 1)

- Run again
- Multi-BLEU score:
  - (my-ph): BLEU = 69.14, 85.2/72.4/64.3/57.6 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.15, 87.8/79.4/75.0/71.3 (BP=1.000, ratio=1.000, hyp\_len=8047, ref\_len=8048)
- NIST-BLEU & NIST-BLEU C are now computed
- No error!

# Run 3 (SMT run 1)

- Error: consonants with athat (်) are not merged
  - Example: line 1600 my\_words= 7 ph\_words= 6
    - my: 'တင်ခဏှပ်ပတ်တိညဏ်'
    - ph: "tin kha nou' pa' ti. njan"
- Dictionary (**final\_syl\_dictionary\_13Feb2024.sorted.txt**) has:
  - ၵ 73
  - ့ 27
  - ...
- Filter those entries, build a new dictionary
- Edit **syl\_normalizer.py** to always return merged results in try\_merge\_with\_previous() function

## Run 4 (SMT run 2)

- Syllable-normalization with the filtered dictionary and the updated **syl\_normalizer.py** on **data/g2p-par/train.my**, **../dev.my**, and **../test.my**
- Create **data/cleaned\_2** for normalized .my files
- Modify paths in **config.baseline2** & Perl scripts
- Generate configs & SGMs
- Run again

# Run 4 (SMT run 2)

- Multi-BLEU score:
  - (my-ph): BLEU = 69.49, 85.2/72.6/64.8/58.2 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.44, 87.8/79.6/75.6/71.7 (BP=1.000, ratio=1.000, hyp\_len=8046, ref\_len=8048)
- No error!

# Run 5 (SMT run 3)

- Use 3-gram instead of 5-gram
- Modify **config.baseline2** and generated configs
  - order = 3
- Run again
- Multi-BLEU score:
  - (my-ph): BLEU = 69.25, 85.3/72.7/64.7/57.4 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.45, 87.9/79.7/75.6/71.7 (BP=1.000, ratio=1.000, hyp\_len=8046, ref\_len=8048)
- No error!

# Run 6 (SMT run 4)

- Lower max-phrase-length
- Modify **config.baseline2** and generated configs
  - max-phrase-length = 3
- Run again
- Multi-BLEU score:
  - (my-ph): BLEU = 69.46, 85.1/72.5/64.2/58.8 (BP=1.000, ratio=1.000, hyp\_len=8051, ref\_len=8048)
  - (ph-my): BLEU = 78.31, 87.9/79.6/75.4/71.4 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)
- No error!

# Run 7 (SMT run 5)

- Less pruning for 3-gram language model
- Modify **config.baseline2** and generated configs
  - settings = "--prune '0 0 0' -T \$working-dir/lm -S 20% --discount\_fallback"
- Run again
- Multi-BLEU score:
  - (my-ph): BLEU = 69.88, 85.3/72.8/65.0/59.1 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.30, 87.7/79.3/75.4/71.8 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)
- No error!

# Run 8 (SMT run 6)

- Install MGIZA & add "**mgizapp, sntcooc, merge\_alignment.py**" to **giza-pp/GIZA++-v2/**
- Modify **config.baseline2** and generated configs
  - training-options = "-mgiza -mgiza-cpus 8"
- Run again
- Multi-BLEU score:
  - (my-ph): BLEU = 69.71, 85.1/72.7/65.0/58.7 (BP=1.000, ratio=1.000, hyp\_len=8050, ref\_len=8048)
  - (ph-my): BLEU = 78.41, 87.8/79.5/75.5/71.8 (BP=1.000, ratio=1.000, hyp\_len=8045, ref\_len=8048)
- No error!

Thank you!!!